Taming Multiparticle Entanglement

Donghun Jung

27 Jan 2026

Department of Physics, Sungkyunkwan University

QuiME Lab, Center for Quantum Technology, Korea

Institute of Science Technology

Paper Information

Taming Multiparticle Entanglement

Authors: Bastian Jungnitsch, Tobias Moroder1, and Otfried Gühne

Journal: Physical Review Letters 106, 190502 (2011)

arXiv: 1010.6049

Key Contributions:

- Efficient criterion for **genuine multipartite entanglement (GME)** via PPT mixtures
- **Semidefinite programming (SDP)** formulation for practical detection
- Fully decomposable entanglement witnesses

Outline

Motivation

The Challenge: How do we efficiently verify that a state has *genuine* multipartite entanglement that cannot be reduced to bipartite correlations?

Separability and Entanglement

A bipartite state ρ is **separable** if:

$$|\psi\rangle = \sum_i \lambda_i |u_i\rangle_A \otimes |v_i\rangle_B$$

$$\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B$$

where $\sum_i p_i = 1$ and $p_i > 0$.

Problem: Directly determining separability is **NP-hard**!

- How can we split the system into two subsystem?
- It is costy operation!

Examples of Multiparticle Entangled States

GHZ State (Greenberger-Horne-Zeilinger)

$$|GHZ_n\rangle = \frac{1}{\sqrt{2}}(|00\cdots0\rangle + |11\cdots1\rangle)$$

W State

$$|W_3\rangle = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle)$$

Cluster State (4-qubit linear)

$$|Cl_4\rangle = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle - |1111\rangle)$$

Examples of Multiparticle Entangled States

For pure state, SVD is the most effective way to quantify entanglement. For example, consider two-qubit bell state (though it is trivial example).

Step 1. Write the state. $|\Phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + 0|01\rangle + 0|10\rangle + \frac{1}{\sqrt{2}}|11\rangle = \sum_{i,j} c_{ij}|i\rangle_A|j\rangle_B$

Step 2. Form Coefficient Matrix.

$$C = \begin{pmatrix} c_{00} & c_{01} \\ c_{10} & c_{11} \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Step 3. Find SVD of C, that is, $C = U\Sigma V^\dagger$.

$$C = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Step 4. So two Schmidt coefficient, suggesting it is entangled state. Try for remaining Bell states!

Challenges for Entanglement Detection and Quantification; pure state

Exponential cost is inevitalbe. For an $m \times n$ matrix:

Resource	Complexity
Time	$\mathcal{O}(\min(mn^2, m^2n)) \simeq \mathcal{O}(mn \cdot \min(m, n))$
Memory	$\mathcal{O}(mn)$ for matrix + $\mathcal{O}(m^2 + n^2)$ for U, V

For N qubits with bipartition $k|(N-k)$, the size of matrix becomes $2^k \times 2^{N-k}$. Thing goes worse if we have to try every possible combination of bipartition, 2^{N-1}.

Challenges for Entanglement Detection and Quantification; mixed state

Even worse, For mixed states, there's no "canonical form" like Schmidt decomposition. The same ρ can be written as:

$$\rho = \sum_k p_k|\psi_k\rangle\langle\psi_k| = \sum_j q_j|\phi_j\rangle\langle\phi_j|$$

with completely different ensembles! One might use only product states (separable), another might not (entangled representation), while the **physical state is the same**.

Biseparable States: Definition

To generalize, a state is separable with respect with to bipartition $M|\bar{M}$, if:

$$\rho^{\text{sep}}_{M|\bar{M}} = \sum_i p_i \, \rho^{(i)}_M \otimes \rho^{(i)}_{\bar{M}}$$

A state is **biseparable** if it can be written as a mixture:

$$\rho^{\text{bs}} = \sum_i p_i \rho^{\text{sep}}_{M_i|\bar{M}_i}$$

where $\sum_i p_i = 1$. Note that non-trivial form, (at least two $p_i \neq 0$), must be mixed sate.

Main Idea: Checking whether PPT mixture exists

It is well-known that separable states are also Positive Partial Transpose (PPT). We denote such states by $\rho^{\text{ppt}}_{M|\bar{M}}$. Thus, we ask whether a state can be written as

$$\rho^{\text{pmix}} = \sum_i p_i \rho^{\text{ppt}}_{M_i|\bar{M}_i}$$

We call states of this form **PPT mixtures**.

Clearly, any biseparable state is a PPT mixture, so proving that a state is no PPT mixture implies **genuine multipartite entanglement**.

Genuine Multipartite Entanglement (GME)

A state is **genuinely multipartite entangled** if and only if it is **NOT biseparable**.

$$\text{GME} = \text{NOT biseparable}$$

Recap: Positive Partial Transpose

For a bipartite state ρ_{AB} with matrix elements $\rho_{ij,kl} = \langle i, j|\rho|k, l\rangle$:

$$\rho^{T_A}_{ij,kl} = \rho_{kj,il}$$

A state has **Positive Partial Transpose (PPT)** if $\rho^{T_A} \geq 0$.

Quick Example: Bell state

$$\rho = |\phi^+\rangle\langle\phi^+| = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \to \rho^{T_A} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Eigenvalues of ρ^{T_A}: $\{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\}$
The **negative eigenvalue** proves the Bell state is entangled!

Strategy: Characterization via entanglement witnesses

Definition

An observable W is an **entanglement witness** for GME if:

- $\mathrm{Tr}(W\rho) \geq 0$ for all biseparable states ρ
- $\mathrm{Tr}(W\rho) < 0$ for some GME state ρ

Decomposable Witnesses

A witness is **decomposable** with respect to bipartition $M|\bar{M}$ if:

$$W = P_M + Q_M^{T_M}$$

where $P_M \geq 0$ and $Q_M \geq 0$.

A witness W is **fully decomposable** if for **all** bipartitions M:

$$W = P_M + Q_M^{T_M}, \quad P_M \geq 0, \quad Q_M \geq 0$$

Fully Decomposable Witnesses

Theorem (Main Result)

Fully decomposable witnesses detect **exactly** the states that are **not PPT mixtures**.

If ρ is not a PPT mixture, then there exists a fully decomposable witness W that detects ρ.

$$\rho \notin \mathcal{S}_{\text{PPT-mix}} \implies \exists\, W \in \mathcal{W}_{\text{fd}} : \mathrm{Tr}(W\rho) < 0$$

where

- $\mathcal{S}_{\text{PPT-mix}}$ = set of PPT mixtures
- $\mathcal{W}_{\text{fd}}$ = set of fully decomposable witnesses
- $\mathrm{Tr}(W\rho) < 0$ = "W detects ϱ"

This suggests that **if we find a fully decomposable W with $\mathrm{Tr}(W\rho) < 0$, the state is GME!**

Example: Fully Decomposable Witnesses

Let's return to Bell state in two-qubit system. There's only one bipartition, $A|B$. The natural candidate for the Bell state is the **projector witness**: $W = \frac{1}{2}I - |\Phi^+\rangle\langle\Phi^+|$.

We need to check if $W = P + Q^{T_A}$ for some $P, Q \geq 0$.

W is decomposable if and only if $W^{T_A} \geq 0$. Computing the partial transpose:

$$(|\Phi^+\rangle\langle\Phi^+|)^{T_A} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \to W^{T_A} = \frac{1}{2}I - (|\Phi^+\rangle\langle\Phi^+||)^{T_A} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The eigenvalues are $\{0, 0, 0, 1\}$. So $W^{T_A} \geq 0$.

Example: Fully Decomposable Witnesses

We can simply choose:

$$P = 0, \qquad Q = W^{T_A}$$

Trivially, we see $\mathrm{Tr} W = 1$, and $P, Q \geq 0$.

Unfortunately, its derivation is non-trivial.

Strategy: SDP Formulation

Semidefinite Program for GME Detection

$$\begin{aligned} \min_{W, P_M, Q_M} \quad & \mathrm{Tr}(W\rho) \\ \text{subject to} \quad & \mathrm{Tr}(W) = 1 \\ & W = P_M + Q_M^{T_M}, P_M \geq 0, Q_M \geq 0 \quad \forall M \end{aligned}$$

The free parameters are given by W and the operators P_M for every subset M . If the minimum of $\mathrm{Tr}(W\rho)$ is negative, ρ is not a PPT mixture and hence **GME**.

The given eqution takes the form of a semidefinite program. The global optimality of an SDP can be certified and the solution can efficiently be computed via interior-point methods.

Additional Strategy:

- fully PPT witnesses; Let $P_M = 0$ so that $W^{T_M} \geq 0$.
- Let $O = \{O_1, \cdots, O_k\}$ be such a set of observables. Then, let $W = \sum_i \lambda_i O_i$.

Numerical Results: White Noise Tolerance

Consider mixed states: $\rho(p) = p \cdot \frac{I}{2^N} + (1-p) \cdot |\psi\rangle\langle\psi|$

Critical noise tolerance p_{tol}: Maximum p where GME is still detected.

State $|\psi\rangle$	**PPT mixtures**	**Previous best**
GHZ_3	**0.571**	0.571
W_3	**0.521**	0.421
$\mathrm{Cluster}_4$	**0.615**	0.533
Dicke_4	**0.539**	0.471
GHZ_4	**0.500**	0.429

Strategy: Modified SDP for Quantification

This approach can also be used to quantify GME. If the trace normalization $\mathrm{Tr}(W) = 1$ is replaced by $0 \leq P_M \leq 1$, and $0 \leq Q_M \leq 1$, the negative witness expectation value is a multipartite entanglement monotone.

$$\begin{aligned} \min_{W, P_M, Q_M} \quad & \mathrm{Tr}(W\rho) \\ \text{subject to} \quad & \mathrm{Tr}(W) = 1 \\ & W = P_M + Q_M^{T_M}, 0 \leq P_M, Q_M \leq I \quad \forall M \end{aligned}$$

Properties of the Monotone $\mathcal{N}(\rho) = -\min \mathrm{Tr}(W\rho)$

- **Vanishes on biseparable states:** $\mathcal{N}(\rho^{\mathrm{bs}}) = 0$
- **Convex:** $\mathcal{N}(\sum_i p_i \rho_i) \leq \sum_i p_i \mathcal{N}(\rho_i)$
- **LOCC monotone:** Doesn't increase under local operations
- **Reduces to negativity** in the bipartite case

PPTMixer: A Python Package for GME Detection

```python
import numpy as np
import cvxpy as cp

def partial_transpose(rho, dims, axis):
    """Compute partial transpose of density matrix."""
    # Reshape, transpose subsystem, reshape back
    ...

def fdecwit(rho, dims):
    """Find optimal fully decomposable witness via SDP."""
    n = rho.shape[0]
    W = cp.Variable((n, n), hermitian=True)
    # Constraints for all bipartitions M
    constraints = [cp.trace(W) == 1]
    for M in bipartitions:
        P_M = cp.Variable((n, n), PSD=True)
        Q_M = cp.Variable((n, n), PSD=True)
        constraints += [W == P_M + partial_transpose(Q_M, dims, M)]

    prob = cp.Problem(cp.Minimize(cp.trace(W @ rho)), constraints)
    return prob.solve()
```

Python Demo: GHZ and W States

State Definitions

```python
def ghz_state(n):
    """n-qubit GHZ state: (|00...0⟩ + |11...1⟩)/√2"""
    d = 2**n
    psi = np.zeros(d, dtype=complex)
    psi[0] = psi[-1] = 1/np.sqrt(2)
    return np.outer(psi, psi.conj())

def w_state():
    """3-qubit W state: (|001⟩ + |010⟩ + |100⟩)/√3"""
    psi = np.array([0, 1, 1, 0, 1, 0, 0, 0], dtype=complex) / np.sqrt(3)
    return np.outer(psi, psi.conj())
```

Python Demo: White Noise Tolerance

Noise Model

$$\rho(p) = p \cdot \frac{\mathbb{1}}{d} + (1-p) \cdot |\psi\rangle\langle\psi|$$

Scanning for Critical p_{tol}

```python
def find_noise_tolerance(pure_state, dims, p_values):
    """Find critical noise level where GME is lost."""
    d = pure_state.shape[0]
    noise = np.eye(d) / d

    for p in p_values:
        rho = p * noise + (1-p) * pure_state
        min_val = fdecwit(rho, dims)
        if min_val >= 0:
            return p  # GME lost at this noise level
    return 1.0
```

Modified SDP with Bounded Operators

```
def entmon(rho, dims):
    """Compute entanglement monotone via bounded SDP."""
    n = rho.shape[0]
    W = cp.Variable((n, n), hermitian=True)
    constraints = [cp.trace(W) == 1]

    for M in bipartitions:
        P_M = cp.Variable((n, n), hermitian=True)
        Q_M = cp.Variable((n, n), hermitian=True)
        # Bounded constraints: 0 ≤ P, Q ≤ I
        constraints += [P_M >> 0, P_M << np.eye(n)]
        constraints += [Q_M >> 0, Q_M << np.eye(n)]
        constraints += [W == P_M + partial_transpose(Q_M, dims, M)]

    prob = cp.Problem(cp.Minimize(cp.trace(W @ rho)), constraints)
    return -prob.solve()  # Return positive value as monotone
```

Summary

References

Main Paper

- Jungnitsch, Bastian, Tobias Moroder, and Otfried Gühne. "Taming multiparticle entanglement." Physical review letters 106.19 (2011): 190502.

Related Works

- Peres, Asher. "Separability criterion for density matrices." Physical Review Letters 77.8 (1996): 1413.
- Horodecki, Michał, Paweł Horodecki, and Ryszard Horodecki. "Separability of n-particle mixed states: necessary and sufficient conditions in terms of linear maps." Physics Letters A 283.1-2 (2001): 1-7.
- Gühne, Otfried, and Géza Tóth. "Entanglement detection." Physics Reports 474.1-6 (2009): 1-75.

Software

- CVXPY: https://www.cvxpy.org/

Thank You!

Questions & Discussion